

Presentation Title

Abstract

We study control policies for satellite-based entanglement distribution in a minimal quantum repeater architecture, consisting of two ground stations and a single satellite equipped with two quantum memories coupled to a single optical interface. The latter property entails that only one elementary down-link can be attempted at a time. In this sequential setting, the first successful link must be stored while the second is generated, creating a tradeoff between waiting for protocol completion and preserving the stored quantum state. Unlike in terrestrial repeater settings, this tradeoff is time dependent during a satellite pass because link success probabilities and communication times vary with orbital geometry. We first compare scheduling rules and find that establishing the first link with the ground station that is further away from the satellite is a favored strategy. We then study policies governing the maximum time a link can be stored, referred to as a cutoff time. To this end, we formulate a time-bin-resolved analytic expression for the secret-key rate that incorporates link probabilities, communication times, memory decoherence, and cutoff-dependent completion probabilities. Using this model, we compare the best orbit-wide static cutoff with dynamic regional cutoff policies. The optimal cutoff varies across the pass, demonstrating that the memory-management problem is intrinsically time dependent. We find that the improvement in total orbit-averaged secret-key rate can be modest, because most key is generated near the center of the pass, where the static cutoff is already close to optimal. Nevertheless, the dynamic cutoffs provide clear advantage at the regional level, especially near the edges of the visibility window. This advantage can be particularly relevant in scenarios where different parts of the orbit service different applications or different users/subscribers at the ground.

1 Motivation

Spend more time on discussing experimental capabilities

- Talk about how satellites came up in quantum networks
 - Optical fibers are lossy
 - Repeaters are needed for long distances, but are hard to implement and expensive
 - Satellites can help distribute entanglement over long distances
- Satellite networks that already/will exist¹

¹Spend more time on discussing experimental capabilities

- SOCRATES
- Micius
- Jinan-1
- Future launches
- Talk about cutoffs and how they're used in current repeater protocols, and how they can be used in satellite networks

2 Main Content

2.1 Past work

Make sure the physical setup is very clear before presenting any results

This is a setup for QKD, we are not trying to generate entanglement between the two ground stations, but rather we are trying to generate a secret key between them. But, explain how, mathematically, purification and bell state measurements are equivalent to entanglement swapping with immediate measurement.

Spend time on setup, explaining what devices and resources are needed

- Explain physical model²³
 - Single downlink from satellite to two ground stations
 - Cutoffs are used to determine when to stop trying to distribute entanglement
 - Explain losses in satellite-to-ground links⁴
 - Our figure of merit is SKR
- Scheduling policies and cutoffs
 - Furthest first is consistently the best
 - There exists a cutoff that maximizes SKR for each policy
 - Dynamic cutoffs can outperform static cutoffs in certain regions of the orbit, but overall they are equivalent
 - Our results are consistent with previous work (Khatri)

2.2 Current work

An extension of our previous work

For this section present an intuition behind why we think the uplink will do best

Just provide intuition behind everything

²Discuss what technology we envision to be on the satellite

³Discuss physical model of the communication scheme before any results

⁴How it varies due to nature of orbits, different for different schemes

- We are comparing our model with other schemes⁵
 - Single downlink with cutoffs (ours), single uplink with cutoffs, double downlink, and double uplink
- We are expanding our model to include more realistic losses and noise
 - Turbulence in the atmosphere, pointing errors, dark counts, and imperfect detectors
 - Decoy states are also used to improve security

⁵Explain visually all of the different schemes, find a way to explain all their differences